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Kelas : 1A - TTE

1.) a. Tekanan memenuhi $P = 10 \log \frac{W}{10^{-12}}$ Jika power W memenuhi $W = \sin 0{,}2t$

hitung laju perubahan tekanan P saat 1 detik?

- $W' = 0{,}2 \cos t$

$P = 10 \log \frac{W}{10^{-12}}$

$P = 10 (\log W - \log 10^{-12})$

$P = 10 (\log W - (-12))$

$P = 10 (\log W + 12)$

$P'(t) = 10 \frac{1}{W \ln 10} \cdot W'$

$P'(1) = 10 \frac{1}{W \ln 10} (0{,}2 \cos t)$

$P'(1) = \frac{10 (0{,}2 \cos t)}{\sin 0{,}2t \ln 10}$

$P'(1) = \frac{10 (0{,}2 \cos (1))}{\sin (0{,}2)(1) \ln 10}$

$P'(1) = \frac{2 \cos 1}{\sin (\frac{1}{5} \times \ln 10)} = 2{,}43 \ N/m^2$

b. Muatan $q(\theta) = 4 \sin 2\theta + 3 \cos 2\theta$ $C \ 0 \leq \theta \leq 2\pi$, $q = 0$ dan $i = 0$ A

$q(\theta) = 4 \sin 2\theta + 3 \cos 2\theta$

$q'(\theta) = 8 \cos 2\theta - 6 \sin 2\theta$

$q'(\theta) = 8 \cos 2\theta - 6 \sin 2\theta$

$4 \cos 2\theta - 3 \sin 2\theta = 0$

$\cot 2\theta = \frac{3}{4}$

$2\theta = \text{arc cot} (3/4)$

$2\theta = \text{arc cot} (3/4) + K.\pi$

$\theta \approx \frac{0{,}463 + K\pi}{2}$

$\theta \approx 26{,}6^\circ + K.90^\circ$

$q = 0$

$q = 4 \sin 2\theta + 3 \cos 2\theta$

$4 \sin 2\theta + 3 \cos 2\theta = 0 \times \cos 2x$

$4 \tan 2\theta = -3$

$\tan 2\theta = -\frac{3}{4}$ → $\theta = \text{arc tan} (-\frac{3}{4})$

$\theta \approx -0{,}321 + \frac{K\pi}{2}$ $\theta \approx -18{,}4^\circ$

Dipindai dengan CamScanner